Université d'Ottawa · University of Ottawa
Faculté des sciences Mathématiques et de statistique
Faculty of Science Mathematics and Statistics

# Introduction to Linear Algebra MAT 1341A

## Midterm 1 - Version A

October 12, 2023

You must **sign below** to confirm that you have read, understand, and will follow these **instructions**:

- This is an 75-minute **closed-book** exam; no notes are allowed. **Calculators are not permitted.** This test has been designed so that a calculator will not be needed.
- The exam consists of 6 questions on 8 pages. Page 8 provides additional work space and you may also use the backs of any of the pages if needed. Indicate clearly where your answer can be found if it is not written down immediately after where the question is given.
- Question 1–3 are short-answer questions. No justification is required for these questions.
- Questions 4–6 are long-answer questions. You must show all relevant steps and give appropriate justifications in order to obtain full marks.
- On Page 7, there is a bonus question. Only attempt it after you have finished all the other questions.
- **Cellular phones** and other electronic devices **are not permitted** during this exam. Phones and other devices must be turned off completely and stored out of reach. Do not keep them in your possession, such as in your pockets. If you are caught with such a device, the following may occur: academic fraud allegations will be filed which may result in your obtaining a 0 (zero) for the exam.

LAST NAME: _____

First name: _____

Student Number: _____

Signature: _____

**Please circle** the DGD section where you would like to have your test returned to you:

| DGD 1 | DGD 2 | DGD 3 | DGD 4 |
|---|---|---|---|
| MON 1:00 – 2:20 | MON 1:00 – 2:20 | MON 5:30 – 6:50 | MON 4:00 – 5:20 |
| STE C0136 | MRT 221 | MRT 256 | MRT 256 |

| Quest. 1 | Quest. 2 | Quest. 3 | Quest. 4 | Quest. 5 | Quest. 6 | Bonus | Total |
|---|---|---|---|---|---|---|---|
| /12 | /5 | /5 | /8 | /10 | /10 | /4 | /50 |

(1) We consider a linear system whose augmented matrix is represented by $B = [A|\mathbf{b}]$, where $A$ is a $10 \times 5$ matrix. Which of the following statements are true? Which are false?

C1. The system has 5 equations and 5 unknowns.

C2. If the rank of $B$ is 5, then the system admits a unique solution.

C3. If $\mathbf{b} = \mathbf{0}$, then the system is inconsistent.

C4. Rank$(B) \geq$Rank$(A)$.

C5. If Rank$(B) >$Rank$(A)$, then the system is consistent.

C6. If Rank$(A) = 3$, then there are infinitely many solutions.

In each box below, circle T if the statement is true or F if the statement if else. [12]

| C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|
| T / F | T / F | T / F | T / F | T / F | T / F |

(2) Let $\mathbb{P}_2$ be the set of polynomials of degree at most 2. Which of the following sets are subspaces of $\mathbb{P}_2$? There may be more than one correct answers.

A. $\{f \in \mathbb{P}_2 : f(1) = 1\}$.

B. $\{f \in \mathbb{P}_2 : f(1) = 0\}$.

C. $\{f \in \mathbb{P}_2 : f(2) = f(3)\}$.

D. $\text{Span}\{X^2 + 1, X^3\}$.

E. $\{f \in \mathbb{P}_2 : f(0) \geq 0\}$.

Answer: B, C [5]

(3) Let $U = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2,2} : a = -b = c \right\}$. Which of the following sets are spanning sets of $U$? There may be more than one correct answers.

A. $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \right\}$.

B. $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

C. $\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

D. $\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\}$.

E. $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

Answer: A, B, E [5]

(4) Suppose a system of linear equations is described by the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & -1 \\ 1 & 4 & 4 & 0 \\ -1 & -1 & e & f \end{array}\right],$$

where $e, f \in \mathbb{R}$ are parameters. What values of $e$ and $f$ cause this system to have no solutions? [8]

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & -1 \\ 1 & 4 & 4 & 0 \\ -1 & -1 & e & f \end{array}\right] \xrightarrow[R_1+R_3 \to R_3]{-R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 2 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 2+e & -1+f \end{array}\right]$$

$$\xrightarrow{-2R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 3 & 2 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & e-2 & -3+f \end{array}\right]$$

So the system is inconsistent if and only if $e-2=0$ and $-3+f \neq 0$

$\iff$ $e=2$ and $f \neq 3$

(5) Let $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

(a) Calculate $A^2, A^3, A^4$. [6]

$$A^2 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & -4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Calculate $A^{2023}$. [4]

The $(1,2)$-entry of $A^n$ is $-n$, the $(3,3)$-entry is $(-1)^n$, while all other entries are unchanged.

So $A^{2023} = \begin{bmatrix} 1 & -2023 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

(6) Let $V = \text{Span}\left\{\begin{bmatrix}1\\1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right\}$.

(a) Determine whether $\begin{bmatrix}1\\2\\-2\\0\end{bmatrix} \in V$. [5]

$$\left[\begin{array}{ccc|c}1&0&0&1\\1&1&0&2\\-1&-1&0&-2\\0&0&1&0\end{array}\right] \xrightarrow{\substack{-R_1+R_2\to R_2\\ R_1+R_3\to R_3}} \left[\begin{array}{ccc|c}1&0&0&1\\0&1&0&1\\0&-1&0&-1\\0&0&1&0\end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c}1&0&0&1\\0&1&0&1\\0&0&0&0\\0&0&1&0\end{array}\right] \xrightarrow{R_3\leftrightarrow R_4} \left[\begin{array}{ccc|c}1&0&0&1\\0&1&0&1\\0&0&1&0\\0&0&0&0\end{array}\right]$$

The system is consistent, so $\begin{bmatrix}1\\2\\-2\\0\end{bmatrix} \in V$.

(b) Show that $\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} \in V$ if and only if $x_2 + x_3 = 0$. [5]

$$\left[\begin{array}{ccc|c}1&0&0&x_1\\1&1&0&x_2\\-1&-1&0&x_3\\0&0&1&x_4\end{array}\right] \sim \left[\begin{array}{ccc|c}1&0&0&x_1\\0&1&0&x_2-x_1\\0&-1&0&x_3+x_1\\0&0&1&x_4\end{array}\right] \sim \left[\begin{array}{ccc|c}1&0&0&x_1\\0&1&0&x_2-x_1\\0&0&0&x_2+x_3\\0&0&1&x_4\end{array}\right] \sim \left[\begin{array}{ccc|c}1&0&0&x_1\\0&1&0&x_2-x_1\\0&0&1&x_4\\0&0&0&x_2+x_3\end{array}\right]$$

The system is consistent if and only if $x_2 + x_3 = 0$

**Bonus:** Let $A$ be an $n \times n$ matrix such that $A^\top A = I_n$. Prove that if $\vec{u}$ and $\vec{v}$ are two distinct columns of $A$, then they are orthogonal to each other. [4]

Write $A = \begin{bmatrix} \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n \end{bmatrix}$

$$A^\top A = \begin{bmatrix} \vec{c}_1^{\,T} \\ \vdots \\ \vec{c}_n^{\,T} \end{bmatrix} \begin{bmatrix} \vec{c}_1 & \cdots & \vec{c}_n \end{bmatrix}$$

$$= \begin{bmatrix} \vec{c}_1 \cdot \vec{c}_1 & \vec{c}_1 \cdot \vec{c}_2 & \cdots & \vec{c}_1 \cdot \vec{c}_n \\ \vec{c}_2 \cdot \vec{c}_1 & \vec{c}_2 \cdot \vec{c}_2 & \cdots & \vec{c}_2 \cdot \vec{c}_n \\ \vdots & & & \\ \vec{c}_n \cdot \vec{c}_1 & \vec{c}_n \cdot \vec{c}_2 & \cdots & \vec{c}_n \cdot \vec{c}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

So $\vec{c}_i \cdot \vec{c}_j = 0 \quad \forall\, i \neq j$

$\Rightarrow \vec{c}_i \perp \vec{c}_j \quad \forall\, i \neq j.$